

Arithmetic Progressions — MCQ Worksheet

Mathematics • Chapter 5 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. An arithmetic progression (AP) is a sequence where:

- (A) Each term differs from previous by fixed number d
- (B) Each term is double the previous
- (C) Each term is square of previous
- (D) No fixed pattern

Q2. The general term of an AP with first term a and common difference d is:

- (A) $a + n d$
- (B) $a + (n - 1) d$
- (C) $a - (n - 1) d$
- (D) $an + d$

Q3. The 10th term of AP: 2, 5, 8, ... is:

- (A) 27
- (B) 29
- (C) 30
- (D) 32

Q4. The 12th term of AP: 7, 11, 15, ... is:

- (A) 47
- (B) 51
- (C) 55
- (D) 43

Q5. The common difference of AP: 3, 8, 13, 18, ... is:

- (A) 3
- (B) 5
- (C) 8
- (D) 13

Q6. In AP: 100, 95, 90, ..., the common difference is:

- (A) 5
- (B) -5
- (C) 100
- (D) 0

Q7. Sum of first n terms of AP is:

- (A) $S_n = n/2[2a + (n-1)d]$
- (B) $S_n = a + (n-1)d$
- (C) $S_n = a \times n$
- (D) $S_n = nd$

Q8. Sum of first 10 natural numbers is:

- (A) 45
- (B) 50
- (C) 55
- (D) 100

Q9. Sum of first 100 natural numbers is:

- (A) 5050
- (B) 5100
- (C) 10000
- (D) 9900

Q10. The number of terms in AP: 7, 13, 19, ..., 205 is:

- (A) 32
- (B) 33
- (C) 34
- (D) 35

Q11. Which term of AP: 21, 18, 15, ... is -81?

- (A) 35th
- (B) 36th
- (C) 34th
- (D) 37th

Q12. If $a_n = 5n - 3$, then $a_5 =$

- (A) 20
- (B) 22
- (C) 24
- (D) 18

Q13. Sum of AP: $2 + 4 + 6 + \dots + 100$ is:

- (A) 2550
- (B) 2500
- (C) 5050
- (D) 5000

Q14. In an AP, if $S_{10} = 100$ and $a = 1$, then $d =$

- (A) 1
- (B) 2
- (C) 3
- (D) $9/5$

Q15. If 7th term is 19 and 12th term is 34, the common difference is:

- (A) 3
(C) 5

- (B) 4
(D) 2

Q16. The sum of n terms of AP $a, a + d, a + 2d, \dots$ if last term is l is:

- (A) $n/2 (a + l)$
(C) $(a + l)/n$

- (B) $n (a + l)$
(D) $n^2(a + l)$

Q17. In an AP, if first term is 5 and last term is 45, sum is 250, then n is:

- (A) 8
(C) 10

- (B) 9
(D) 11

Q18. a, b, c are in AP iff:

- (A) $b^2 = ac$
(C) $a + b = c$

- (B) $2b = a + c$
(D) $a = b + c$

Q19. Three numbers in AP whose sum is 27 and product is 648 (smallest) are:

- (A) 6, 9, 12
(C) 1, 9, 17

- (B) 3, 9, 15
(D) 6, 8, 13

Q20. The common difference of AP whose n th term is $2n + 5$ is:

- (A) 2
(C) 7

- (B) 5
(D) $2n$

Q21. The 50th term of AP: 5, 9, 13, ... is:

- (A) 205
(C) 209

- (B) 201
(D) 197

Q22. Sum to 20 terms of AP: 1, 4, 7, ... is:

- (A) 590
(C) $590/2$

- (B) 580
(D) 620

Q23. AP whose first term is -3 and common difference is 2: 5th term is:

- (A) 3
(C) 7

- (B) 5
(D) 9

Q24. How many two-digit numbers are divisible by 7?

- (A) 12
(C) 14

- (B) 13
(D) 15

Q25. Sum of first 25 odd natural numbers is:

- (A) 625
(C) 675

- (B) 525
(D) 725